- Batch – B5

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# EEC23702 -Mini Project Work

## EV-Agent: Agentic AI Powered Smart EV Charging Platform

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Final review on 28.09.26

# Outline of the Presentation

- Objective / Objectives
- Literature Survey
- Existing System
- Scope of the Project
- Proposed System
- System Architecture
- Methodology / Workflow
- Technologies & Implementation
- Result & Output
- Conclusion & Future Scope
- References

# Objectives

- **Optimize EV Charging** – Determine the best time and method to charge an electric vehicle efficiently.
- **Find Available Charging Stations** – Recommend suitable nearby charging stations based on availability and user requirements.
- **Reduce Charging Cost** – Suggest cost-effective charging schedules by considering electricity prices and demand.

# Literature Survey

| Year | Title of The Research Paper | Author | Methodology | Existing System | Drawbacks |
|---|---|---|---|---|---|
| 2021 | EV charging infrastructure in India | Mishra et al. | Reviewed development, infrastructure requirements and challenges of EV charging in India | EV charging infrastructure is mainly focused on deployment of charging stations, infrastructure development, and basic charging services. | No intelligent AI-based user recommendation |
| 2022 | Optimal EV charging-station location | Ahmad et al. | Reviewed station placement, charging demand, optimization and grid impact | Existing systems focus on optimal charging-station placement, demand estimation, and grid-aware planning using location and charging-demand data. | Focuses mainly on infrastructure planning rather than real-time user assistance |

# Literature Survey

| Year | Title of The Research Paper | Author | Methodology | Existing System | Drawbacks |
|---|---|---|---|---|---|
| 2023 | **Charging technologies & station recommendation** | **Charging Technology & Recommendation Review** | Reviewed charging technologies, infrastructure and station-finding/recommendation methods | Existing systems provide charging-station recommendation based on station location, availability, charging requirements, and user preferences. | Does not provide an Agentic AI system capable of reasoning and taking actions |

# Existing System & Problems

## Existing System

- Manual Charging : Users manually select when and where to charge.
- Basic Charging Apps : Show charging stations and their availability.
- Fixed Scheduling : Charging is scheduled using predefined time slots.
- Limited AI : Existing systems provide basic recommendations.

## Problems

- High charging cost during peak hours.
- Charging stations may be unavailable when users arrive.
- No personalized charging plans.
- Limited real-time decision-making.

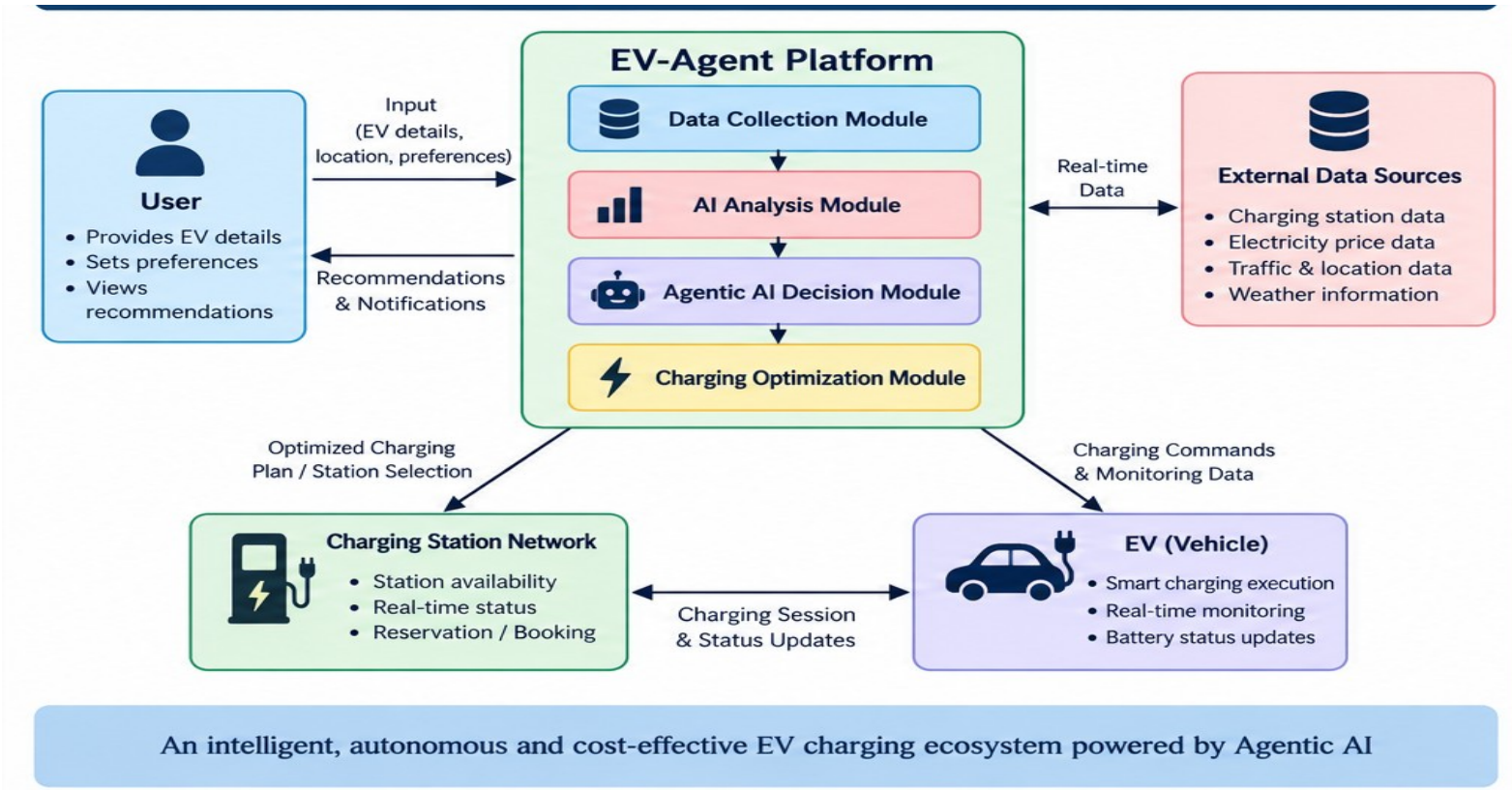
# Scope of the Project

- **Smart Charging** – Optimize charging time based on battery level and user needs.
- **Station Selection** – Identify suitable charging stations based on location and availability.
- **Cost Optimization** – Recommend cost-effective charging options.
- **Agentic AI** – Enable autonomous and intelligent charging decisions.
- **Personalized Planning** – Adapt charging plans according to user preferences.
- **Energy Efficiency** – Promote efficient energy usage and reduce peak demand.

# Proposed System

The proposed system begins by collecting user and EV details, including battery status and charging requirements. It then performs battery status analysis and collects relevant charging station data to understand available charging options. Based on the user's requirements, the system performs charging demand prediction and identifies the most suitable charging station and time. An Agentic AI decision-making module evaluates these factors along with cost and energy consumption to determine an optimized charging strategy. Finally, the system provides a personalized charging recommendation and continuously monitors the charging process through smart charging and real-time monitoring.

# System Architecture



# Methodology / Workflow

The proposed system consists of six sequential steps, starting from EV and user data input and ending with personalized smart charging recommendations.

**1. User & EV Data Input**

Input Collection: User details, EV battery capacity, current State of Charge (SoC), vehicle range, destination, and preferred charging requirements are collected as input to the system.

**2. Battery & Range Analysis**

Battery Analysis: The system analyzes the current battery status, remaining range, estimated energy requirement, and expected consumption to determine whether charging is required.

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**3. Charging Station Data Collection**

Station Data: Real-time charging station information such as location, availability, charger type, charging speed, distance, pricing, and operating status is collected.

**4. Charging Demand Prediction**

Demand Prediction: Machine learning models analyze battery status, travel distance, historical charging patterns, and station demand to estimate the required charging demand and suitable charging time.

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**5. Agentic AI Optimization**

Intelligent Decision Making: The Agentic AI evaluates multiple factors such as distance, charging time, station availability, electricity cost, energy consumption, and user preferences to select an optimized charging station and charging schedule.

**6. Personalized Recommendation & Monitoring**

Smart Recommendation: The system provides a personalized charging recommendation, including the recommended station, charging time, estimated cost, and expected energy requirement, while continuously monitoring charging conditions and updating the recommendation when required.

# Technologies & Implementation

The proposed EV Charging Recommendation system is implemented using Python with libraries such as Pandas, NumPy, Scikit-learn, Matplotlib, and Seaborn. The system uses EV and charging-station datasets containing information such as battery capacity, State of Charge (SoC), vehicle range, charging demand, station availability, charging speed, distance, and charging cost. Pandas and NumPy are used for data preprocessing, cleaning, and numerical analysis, while machine learning techniques are used to predict charging demand and estimate EV requirements. The system integrates real-time charging-station information to evaluate available stations based on distance, availability, charging time, and cost.

# Technologies & Implementation

## Software / Library / Module Used:

- **Programming Language:** Python
- **Libraries:** Pandas, NumPy, Scikit-learn
- **Machine Learning Model**: Random Forest Regressor for charging-demand prediction
- **Prediction Inputs:** Battery SoC, time, station usage, travel distance, and weather conditions
- **Data Processing**: Pandas and NumPy are used for data cleaning, preprocessing, and numerical analysis
- **Evaluation Metrics:** MAE, RMSE, and $R^2$ are used to evaluate prediction performance

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## Software / Library / Module Used:

- **Mapping & Routing:** OpenStreetMap and OSRM are used for location, distance, and route analysis
- **Charging Station Data:** Open Charge Map is used to obtain charging-station information and availability
- **Weather Data**: Open-Meteo is used to incorporate weather conditions into charging-demand analysis
- **Agentic AI:** Used for intelligent decision-making, station selection, cost and energy optimization, and personalized charging recommendations
- **Backend & Database:** Flask with SQLite/MySQL for API services, data management, and system integration

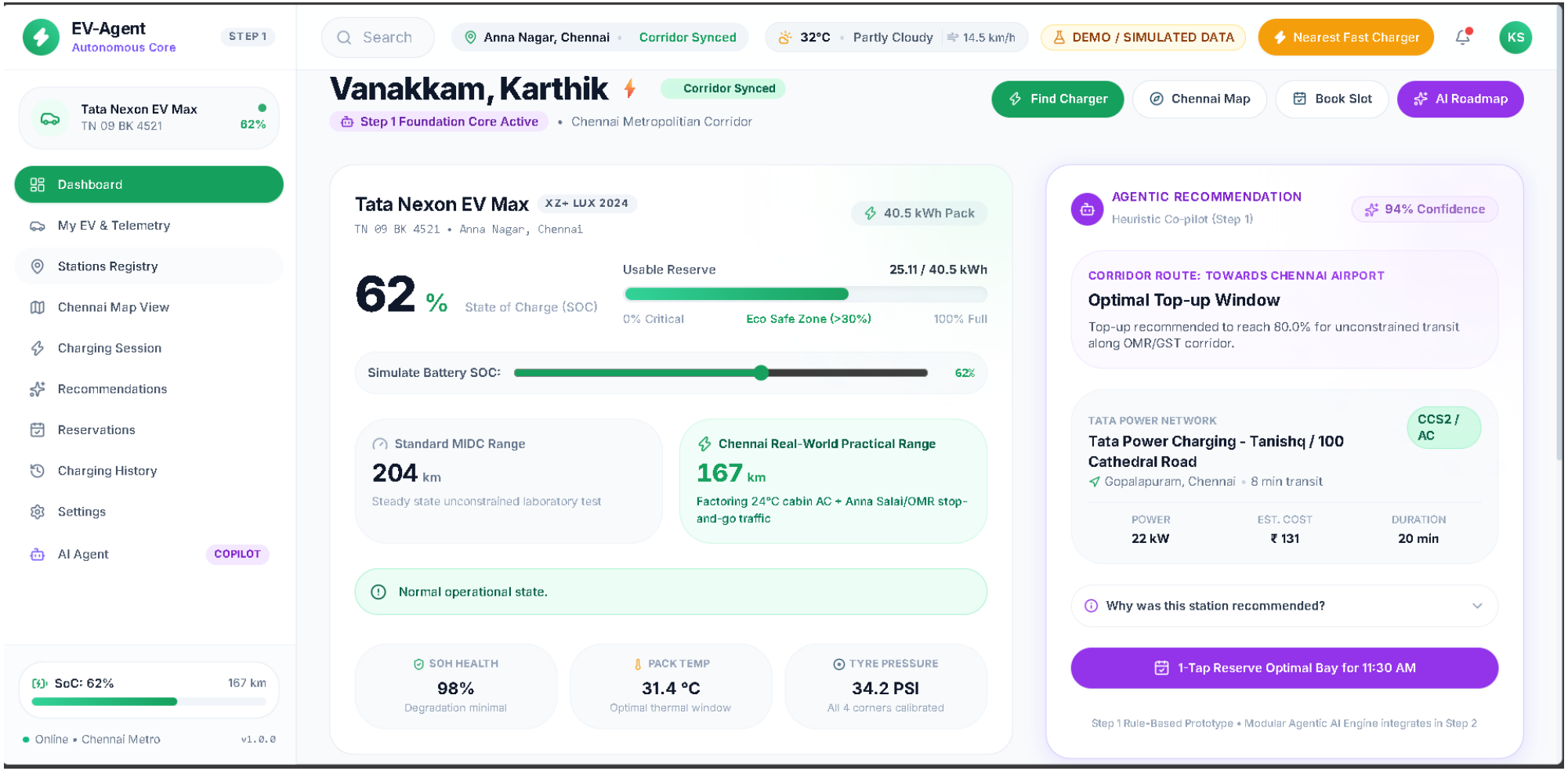
# Results

- Successfully collected and preprocessed EV and charging-station data for analysis and recommendation.
- Predicted charging demand using battery status, travel distance, charging patterns, and other relevant parameters.
- Identified suitable charging stations based on distance, charger availability, charging speed, and station status.
- Optimized charging station and charging time by considering user requirements, estimated charging duration, cost, and energy consumption.
- Agentic AI generated personalized charging recommendations by evaluating multiple factors and selecting an appropriate charging strategy.

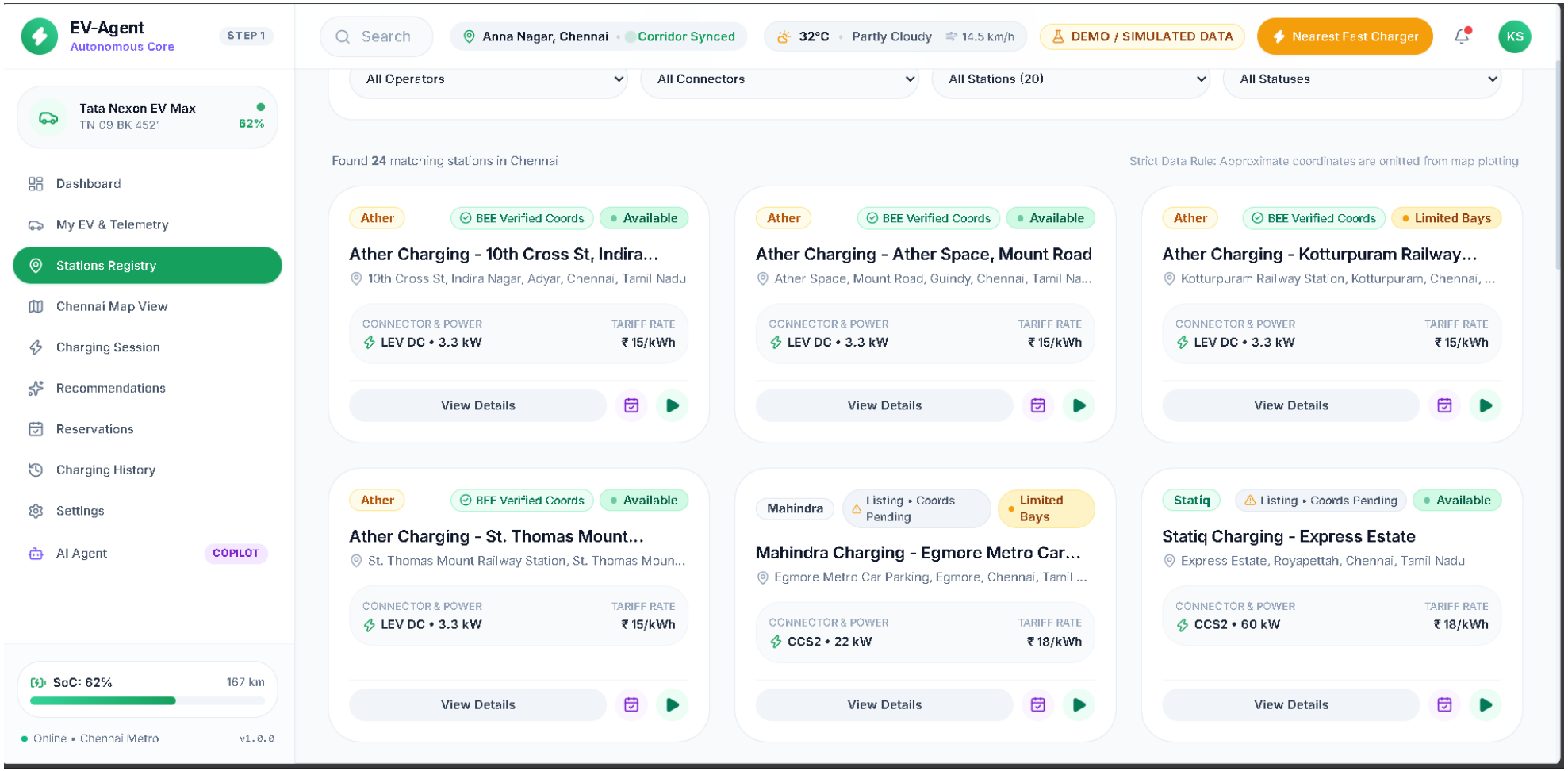
# Conti..

- The system demonstrates the complete workflow: EV & User Input → Battery Analysis → Station Data → Demand Prediction → AI Optimization → Personalized Recommendation → Real-Time Monitoring.
- The proposed system reduces manual decision-making by providing intelligent, data-driven recommendations for efficient and convenient EV charging.

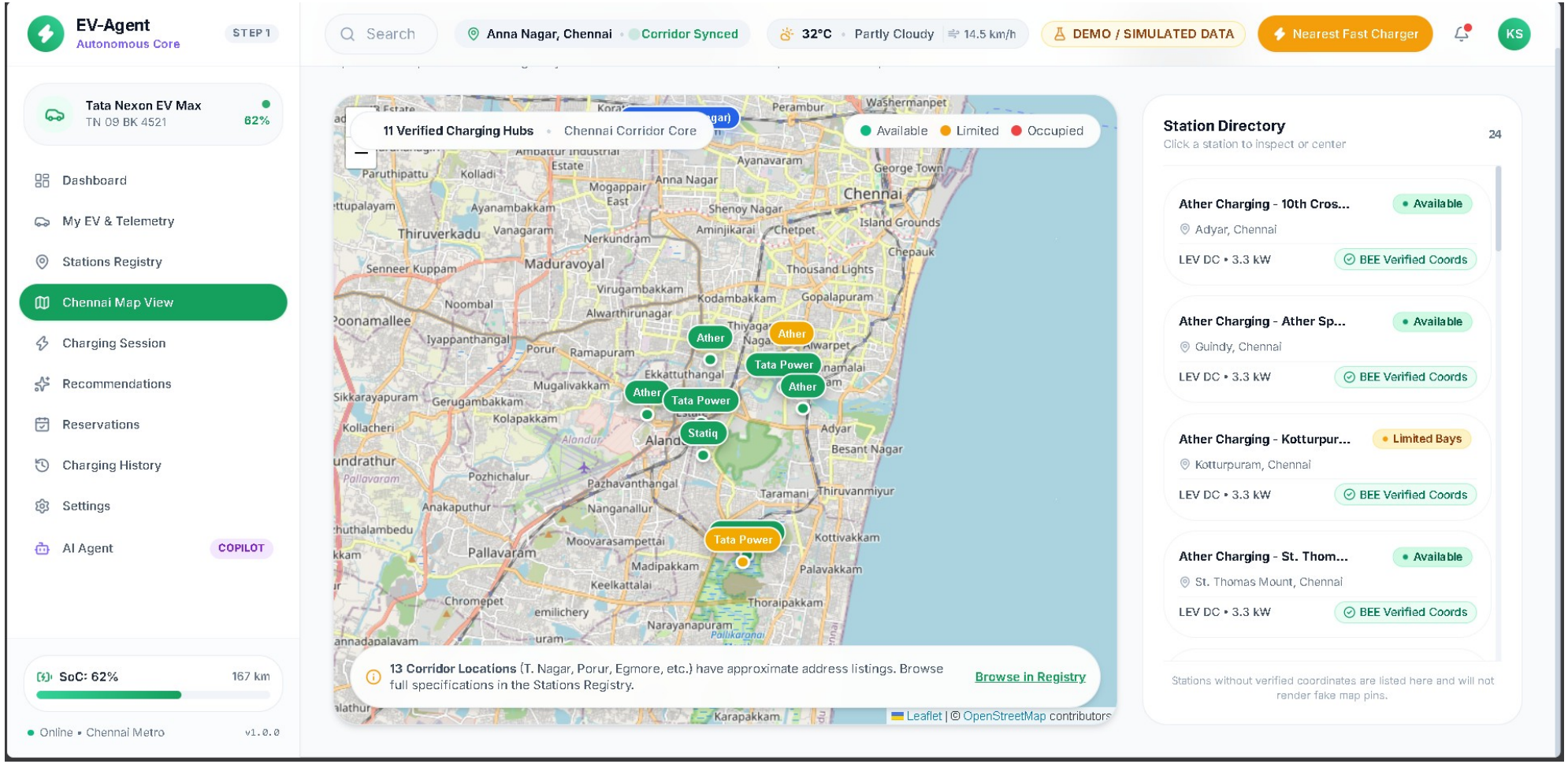
# Output(Screen Shot)



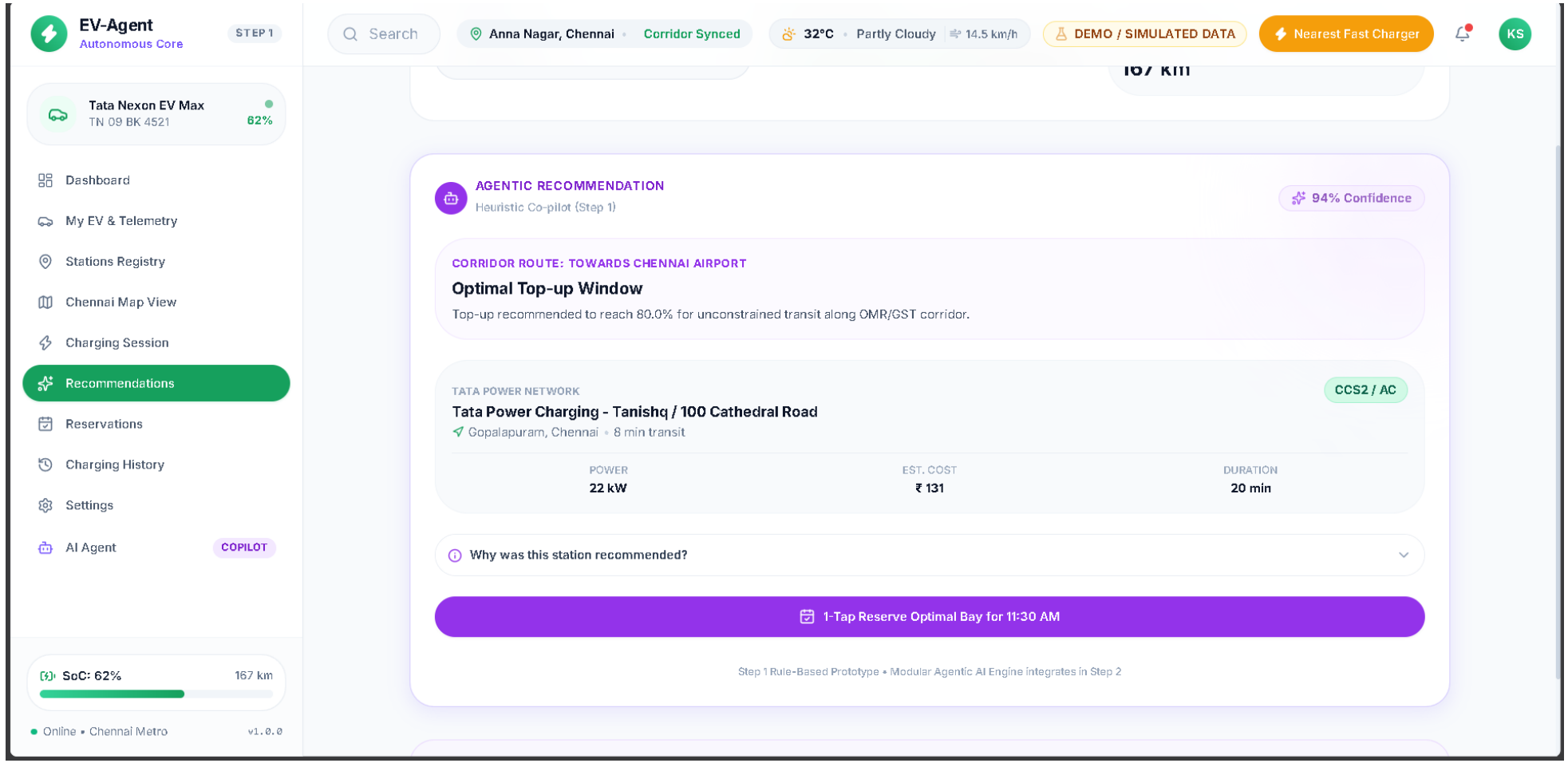
# Conti..


EV-Agent
Autonomous Core
STEP 1
Search
Anna Nagar, Chennai
Corridor Synced
32°C
Partly Cloudy
14.5 km/h
DEMO / SIMULATED DATA
Nearest Fast Charger
KS
Tata Nexon EV Max
TN 09 BK 4521
62%
Dashboard
My EV & Telemetry
Stations Registry
Chennai Map View
Charging Session
Recommendations
Reservations
Charging History
Settings
AI Agent
COPILOT
SoC: 62%
167 km
Online • Chennai Metro
v1.0.0
All Operators
All Connectors
All Stations (20)
All Statuses
Found 24 matching stations in Chennai
Strict Data Rule: Approximate coordinates are omitted from map plotting
Ather
BEE Verified Coords
Available
Ather Charging - 10th Cross St, Indira...
10th Cross St, Indira Nagar, Adyar, Chennai, Tamil Nadu
CONNECTOR & POWER
LEV DC • 3.3 kW
TARIFF RATE
₹ 15/kWh
View Details
Ather
BEE Verified Coords
Available
Ather Charging - Ather Space, Mount Road
Ather Space, Mount Road, Guindy, Chennai, Tamil Na...
CONNECTOR & POWER
LEV DC • 3.3 kW
TARIFF RATE
₹ 15/kWh
View Details
Ather
BEE Verified Coords
Limited Bays
Ather Charging - Kotturpuram Railway...
Kotturpuram Railway Station, Kotturpuram, Chennai, ...
CONNECTOR & POWER
LEV DC • 3.3 kW
TARIFF RATE
₹ 15/kWh
View Details
Ather
BEE Verified Coords
Available
Ather Charging - St. Thomas Mount...
St. Thomas Mount Railway Station, St. Thomas Moun...
CONNECTOR & POWER
LEV DC • 3.3 kW
TARIFF RATE
₹ 15/kWh
View Details
Mahindra
Listing • Coords Pending
Limited Bays
Mahindra Charging - Egmore Metro Car...
Egmore Metro Car Parking, Egmore, Chennai, Tamil ...
CONNECTOR & POWER
CCS2 • 22 kW
TARIFF RATE
₹ 18/kWh
View Details
Statiq
Listing • Coords Pending
Available
Statiq Charging - Express Estate
Express Estate, Royapettah, Chennai, Tamil Nadu
CONNECTOR & POWER
CCS2 • 60 kW
TARIFF RATE
₹ 18/kWh
View Details

# Conti..


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Corridor Synced
32°C
Partly Cloudy
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Charging History
Settings
AI Agent
COPILOT
SoC: 62%
167 km
Online • Chennai Metro
v1.0.0
11 Verified Charging Hubs
Chennai Corridor Core
Available
Limited
Occupied
Ather
Tata Power
Statiq
13 Corridor Locations (T. Nagar, Porur, Egmore, etc.) have approximate address listings. Browse full specifications in the Stations Registry.
Browse in Registry
Leaflet | © OpenStreetMap contributors
Station Directory
Click a station to inspect or center
24
Ather Charging - 10th Cros...
Available
Adyar, Chennai
LEV DC • 3.3 kW
BEE Verified Coords
Ather Charging - Ather Sp...
Available
Guindy, Chennai
LEV DC • 3.3 kW
BEE Verified Coords
Ather Charging - Kotturpur...
Limited Bays
Kotturpuram, Chennai
LEV DC • 3.3 kW
BEE Verified Coords
Ather Charging - St. Thom...
Available
St. Thomas Mount, Chennai
LEV DC • 3.3 kW
BEE Verified Coords
Stations without verified coordinates are listed here and will not render fake map pins.

# Conti..


EV-Agent
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STEP 1
Search
Anna Nagar, Chennai
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32°C
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Charging Session
Recommendations
Reservations
Charging History
Settings
AI Agent
COPILOT
SoC: 62%
167 km
Online • Chennai Metro
v1.0.0
167 KM
AGENTIC RECOMMENDATION
Heuristic Co-pilot (Step 1)
94% Confidence
CORRIDOR ROUTE: TOWARDS CHENNAI AIRPORT
Optimal Top-up Window
Top-up recommended to reach 80.0% for unconstrained transit along OMR/GST corridor.
TATA POWER NETWORK
CCS2 / AC
Tata Power Charging - Tanishq / 100 Cathedral Road
Gopalapuram, Chennai • 8 min transit
POWER
22 kW
EST. COST
₹ 131
DURATION
20 min
Why was this station recommended?
1-Tap Reserve Optimal Bay for 11:30 AM
Step 1 Rule-Based Prototype • Modular Agentic AI Engine integrates in Step 2

# Conclusion

- The proposed EV Charging Recommendation System provides an intelligent and personalized approach to EV charging by combining battery analysis, charging-station data, demand prediction, and optimization techniques.

- The system analyzes user requirements, battery status, station availability, distance, charging time, cost, and energy consumption to identify suitable charging options.

- The integration of Agentic AI enables the system to make context-aware decisions and generate personalized charging recommendations instead of relying only on fixed rules.

# Future Scope

- Real-Time EV Integration: Integrate live vehicle data such as battery SoC, energy consumption, and remaining range directly from the EV.

- Advanced Demand Prediction: Use advanced Machine Learning and Deep Learning models to improve charging-demand and energy-consumption predictions.

- Dynamic Pricing Optimization: Incorporate real-time electricity tariffs and charging-station pricing to identify cost-efficient charging schedules.

- V2G Integration: Extend the system to support Vehicle-to-Grid (V2G) technology, allowing EVs to supply stored energy back to the grid when required.

# Future Scope

- Multi-Agent AI: Develop multiple specialized AI agents for route planning, station selection, energy optimization, and cost management.

- Smart Grid Integration: Connect the system with smart-grid infrastructure to balance charging demand and reduce peak-load conditions.

- Mobile Application: Develop a mobile application that provides live station availability, navigation, charging recommendations, and notifications.

- Predictive Maintenance: Use historical charging-station data to predict equipment failures and improve station reliability.

# References

1. International Energy Agency (IEA), "Global EV Outlook 2025," International Energy Agency.

2. U.S. Department of Energy, Alternative Fuels Data Center (AFDC), Electric Vehicle Charging Infrastructure.

3. Open Charge Map, Global Electric Vehicle Charging Locations and Data.

4. OpenStreetMap Foundation, OpenStreetMap: Open Geographic Database.

5. OSRM, Open Source Routing Machine, Route and Distance Services.

6. Open-Meteo, Weather API and Forecast Data.

# PATENT/APPLICATION

- **EVChargeAI is a mobile application for smart and personalized EV charging recommendations.**
- Collects EV details, battery SoC, location, destination, and charging preferences.
- Analyzes nearby charging stations, availability, distance, charging speed, and cost.
- Uses Machine Learning and Agentic AI to predict charging demand and recommend the best-suited station and charging time.
- Provides navigation, estimated charging cost, and real-time charging monitoring.

# Thank You!

For your attention

